

# Multimodal Pedagogical Friction Detector: Progress Report

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## Abstract

Effective teaching depends on noticing student confusion and adjusting instruction. We are building a system that detects *pedagogical friction*: moments where a low-quality instructional strategy coincides with visible student confusion. The pipeline takes a classroom or tutoring recording (video + audio), constructs a confusion timeline from facial affect (DeepFace), a strategy timeline from automatically transcribed speech (Whisper) and dialogue-level heuristics standing in for Edu-ConvoKit and Tutor CoPilot classifiers, aligns both streams on a 30-second grid, and applies a heuristic pre-filter before optional multimodal fusion with Qwen2.5-VL (replacing GPT-4o in our proposal) to produce a post-session *Teacher Friction Report*. We have implemented the end-to-end codebase, evaluated TIMSS-style public lessons locally, and run CPU smoke tests. Full VLM fusion and quantitative evaluation against human labels are in progress.

## 1 Introduction

Pedagogical friction bridges *what the teacher says* and *how students appear to respond*. Prior work such as Tutor CoPilot improves tutor strategies using text [?], and Edu-ConvoKit supports analysis of teacher discourse [?], but classroom video contains complementary evidence (facial affect, board work) that text-only tools miss. Our goal is to surface candidate friction intervals with timestamps, strategy context, and suggested higher-quality moves, grounded in accountable-talk style strategy taxonomies.

## 2 Related Work

**Classroom discourse and feedback.** Talk Moves and related frameworks motivate labeling instructional moves [?]. **Tutor and dialogue modeling.** Tutor CoPilot and Edu-ConvoKit provide models and tooling we plan to integrate for production-quality strategy labels [?, ?]. **Multimodal reasoning.** Vision-language models such as Qwen2.5-VL support image-text reasoning [?]; we use them for fusion instead of GPT-4o for cost and reproducibility. **Affect from video.** Off-the-shelf emotion models (e.g., via DeepFace) offer a practical confusion proxy on CPU.

## 3 Methodology

Figure ?? summarizes our pipeline (conceptually matching the proposal). **Vision stream:** frames sampled at configurable FPS (default 2.0; reduced for CPU smoke tests); DeepFace emotion → per-frame confusion proxy; trailing 30s mean → confusion timeline. **Language stream:** Whisper (faster-whisper) transcription; utterances annotated with a *heuristic* layer for low-quality / high-pressure patterns, with hooks to replace

by Edu-ConvoKit + Tutor CoPilot. **Alignment and filtering:** 30s bins; flag bins where confusion exceeds  $\mu + z\sigma$  and strategy is low-quality or high-pressure. **Fusion (optional):** Qwen2.5-VL receives frame crops + transcript excerpt + timeline summaries and outputs JSON: friction (bool), rationale, alternative strategy. **Stretch (not yet integrated end-to-end):** OCR on board crops (Tesseract / vision OCR).

*[Insert pipeline figure: Video+Audio  $\rightarrow$  DeepFace + Whisper  $\rightarrow$  Heuristic strategies  $\rightarrow$  Grid + filter  $\rightarrow$  Qwen2.5-VL  $\rightarrow$  Teacher Friction Report]*

Figure 1: Schematic of our system (replace this placeholder with a PowerPoint/Keynote export PDF for crisp, selectable text).

| Setting                                     | Precision | Recall | Notes                      |
|---------------------------------------------|-----------|--------|----------------------------|
| Heuristic candidates + fusion (full lesson) | XX        | XX     | Pending GPU run / labels   |
| Heuristic candidates only (–skip-fusion)    | –         | –      | Smoke tests on TIMSS clips |
| Human agreement (2 annotators)              | XX        | –      | Planned                    |

Table 1: Preliminary experimental plan. **XX** denotes pending results; we will report point-level agreement and friction-point PR against dual-annotated bins.

## 4 Experimental Settings

**Data.** Public TIMSS Video lessons (YouTube mirrors) and transcript bundles from `timssvideo.com`; repository includes transcript archives; video files remain local (`.gitignore`). **Software.** Python 3.9 CPU path: `requirements-pipeline.txt` (OpenCV, DeepFace, TensorFlow, faster-whisper, imageio-ffmpeg); Qwen path requires Python  $\geq 3.10$  and `requirements.txt`. **Reproducibility.** Code: <https://github.com/jw310-wjr/comp646-friction-detector>. **Runtime.** Full 40+ minute lessons at 2 FPS imply thousands of DeepFace calls on CPU; we support `--max-duration-sec` for trimmed evaluation.

## 5 Analysis and Discussion

### 5.1 Implementation status

We completed frame extraction, confusion aggregation, ASR, bin alignment, JSON/text reports, and GitHub distribution. Short clips sometimes yield zero heuristic candidates under conservative  $z$  thresholds; tuning  $z$ , FPS, and heuristic coverage will be reported in the final write-up.

## 5.2 Fusion and compute

Qwen2.5-VL integration is implemented but was not exercised on the course laptop under Python 3.9; we will run fusion on a 3.10+ environment with GPU.

## 5.3 Limitations

Heuristic strategy tags are not yet interchangeable with published Edu-ConvoKit / Tutor CoPilot checkpoints. Friction labels for evaluation still need collection.

## 6 Conclusion

We implemented a working multimodal friction-detection prototype with public-facing code and TIMSS-aligned text resources. Next steps are VLM fusion at scale, classifier replacement, human annotation, and quantitative metrics from Section 3.

## 7 Ethical and Societal Considerations

Classroom video involves minors and privacy: we use only publicly released TIMSS public-use lessons and follow site terms. Automated affect inference can misread culture, disability, or camera angle; reports are meant for teacher reflection, not high-stakes judgment. Mitigations include human review, transparent heuristics, and clear disclaimers in the UI/report.

## Acknowledgments

We used GitHub Copilot / LLM assistants for coding and drafting help; all technical decisions and claims remain our responsibility. We thank the TIMSS Video and TalkMoves communities for public materials.

## References

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